

BAB 14 - SharedPreferences

A. Membuat Tabel User

1. Buatlah sebuah tabel bernama **user** didalam database **kampus** dengan struktur tabel sebagai berikut :

Nama tabel : user				
Nama Kolom	Tipe Data	Panjang	Atribut	Keterangan
user_name	VARCHAR	50	UNIQUE, NOT NULL	Menyimpan username secara unik
pass_word	VARCHAR	50	NOT NULL	Menyimpan password
id	INT		PRIMARY KEY, AUTO INCREMENT, NOT NULL	Menyimpan ID user

2. Tambahkan data ke dalam tabel **user**, semisal, **user_name=user@gmail.com**, **pass_word=123abc** dengan menerapkan *hashing* MD5 kedalam kolom **pass_word** sehingga password menjadi tidak mudah untuk dibaca. Contoh untuk query insert dan select menjadi seperti berikut ini :

- *Insert*

```
INSERT INTO kampus.user(user_name,pass_word)
VALUES ('user@gmail.com',MD5('123abc'));
```

- *Select*

```
SELECT * FROM kampus.user
WHERE user_name='user@gmail.com' AND pass_word=MD5('123abc');
```

Yang akan memberikan hasil sebagai berikut

user_name	pass_word	id
user@gmail.com	a906449d5769fa7361d7ecc6aa3f6d28	2

B. Membuat Web API untuk Login

1. Buatlah file php dengan nama **login.php** dan isikan kode program PHP seperti berikut ini

```
<?php
header("Content-Type: application/json; charset=UTF-8");
header("Access-Control-Allow-Origin: *");
header("Access-Control-Allow-Methods: POST");

$host      = "localhost";
$database  = "kampus";
$username_db = "root";
$password_db = "";

try {
    $pdo = new PDO("mysql:host=$host;dbname=$database", $username_db,
        $password_db);
    $pdo->setAttribute(PDO::ATTR_ERRMODE, PDO::ERRMODE_EXCEPTION);
} catch (PDOException $e) {
    echo json_encode(['sukses' => '111']);
    exit();
}
```

```

}

$username = isset($_POST['username']) ? $_POST['username'] : '';
$password = isset($_POST['password']) ? $_POST['password'] : '';
$result = ['sukses' => '111'];

if (!empty($username) && !empty($password)) {
    try {
        $sql = "SELECT id, user_name, pass_word FROM user WHERE user_name =
                :username AND pass_word = MD5(:password)";

        $stmt = $pdo->prepare($sql);
        $stmt->bindParam(':username', $username);
        $stmt->bindParam(':password', $password);
        $stmt->execute();

        $row = $stmt->fetch(PDO::FETCH_ASSOC);
        if (count($row) > 0) {
            $result = ['sukses' => '000'];
        }
    } catch (PDOException $e) {
        $result = ['sukses' => '111'];
    }
}

echo json_encode($result);
?>

```

C. Membuat SharedPreferences untuk Login dan Dashboard

1. SharedPreferences untuk Login ini bertujuan untuk memudahkan dalam proses login ke aplikasi Android dengan menyimpan data login yang benar ke SharedPreferences sehingga user tidak perlu mengisi username/password lagi setiap kali akan login ke aplikasi

2. Buatlah *project* baru dengan detail sebagai berikut:

Application Name : App 14
Package Name : nugroho.benni.app14 (sesuaikan dengan nama anda)
Minimum SDK : API 25

3. Pada file AndroidManifest.xml tambahkan **Permission Internet**, dan **usesCleartextTraffic**.
Atur juga untuk **theme** aplikasi gunakan **Theme.AppCompat.Light**

```

<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">
    <uses-permission
        android:name="android.permission.INTERNET"></uses-permission>

    <application
        android:usesCleartextTraffic="true"
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"

```

```

        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportRtl="true"
        android:theme="@style/Theme.AppCompat.Light"
        tools:targetApi="31">
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />
                <category
                    android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
        <activity android:name=".DashboardActivity"></activity>
    </application>
</manifest>

```

4. Pada build.gradle (module) tambahkan pustaka Volley seperti berikut

```

plugins {
    alias(libs.plugins.android.application)
    alias(libs.plugins.kotlin.android)
}

android {
    namespace = "nugroho.benni.app14"
    compileSdk = 35

    defaultConfig {
        applicationId = "nugroho.benni.app14"
        minSdk = 25
        targetSdk = 35
        versionCode = 1
        versionName = "1.0"

        testInstrumentationRunner =
            "androidx.test.runner.AndroidJUnitRunner"
    }

    buildTypes {
        release {
            isMinifyEnabled = false
            proguardFiles(
                getDefaultProguardFile("proguard-android-optimize.txt"),
                "proguard-rules.pro"
            )
        }
    }

    buildFeatures{
        viewBinding = true
    }

    compileOptions {
        sourceCompatibility = JavaVersion.VERSION_11
        targetCompatibility = JavaVersion.VERSION_11
    }
}

```

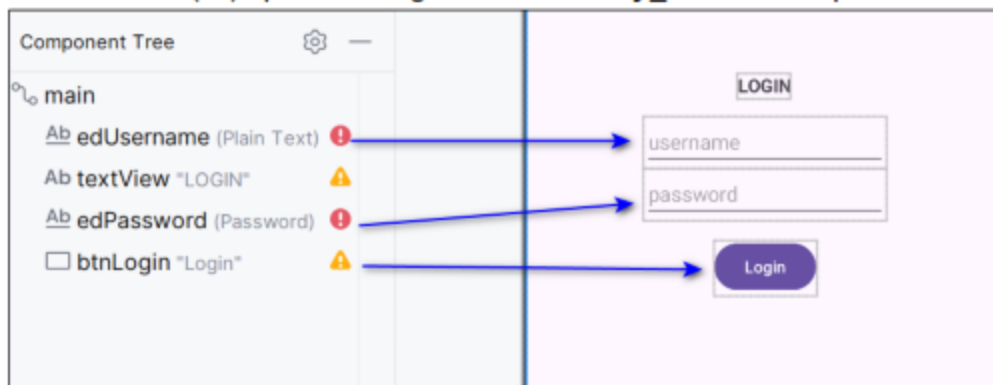
```

kotlinOptions {
    jvmTarget = "11"
}

dependencies {
    implementation("com.android.volley:volley:1.2.1")
    implementation(libs.androidx.core.ktx)
    implementation(libs.androidx.appcompat)
    implementation(libs.material)
    implementation(libs.androidx.activity)
    implementation(libs.androidx.constraintlayout)
    testImplementation(libs.junit)
    androidTestImplementation(libs.androidx.junit)
    androidTestImplementation(libs.androidx.espresso.core)
}

```

5. Buat *layout* antarmuka (UI) aplikasi dengan nama **activity_main.xml** seperti berikut



Kode XML untuk antarmuka dalam **activity_main.xml** sebagai berikut

```

<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <EditText
        android:id="@+id/edUsername"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="16dp"
        android:ems="10"
        android:hint="username"
        android:inputType="text"
        app:layout_constraintBottom_toTopOf="@+id/edPassword"
        app:layout_constraintEnd_toEndOf="@+id/textView"
        app:layout_constraintStart_toStartOf="@+id/textView"
        app:layout_constraintTop_toBottomOf="@+id/textView" />

    <TextView
        android:id="@+id/textView"
        android:layout_width="wrap_content"

```

```

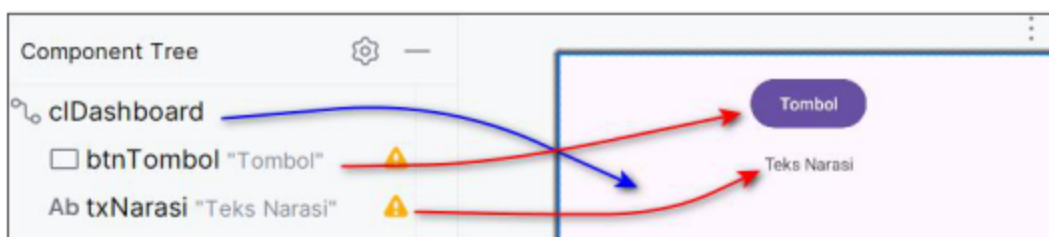
        android:layout_height="wrap_content"
        android:text="LOGIN"
        android:textSize="16sp"
        android:textStyle="bold"
        app:layout_constraintBottom_toTopOf="@+id/edUsername"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.5"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_chainStyle="packed" />

<EditText
    android:id="@+id/edPassword"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:ems="10"
    android:hint="password"
    android:inputType="textPassword"
    app:layout_constraintBottom_toTopOf="@+id/btnLogin"
    app:layout_constraintEnd_toEndOf="@+id/edUsername"
    app:layout_constraintHorizontal_bias="0.5"
    app:layout_constraintStart_toStartOf="@+id/edUsername"
    app:layout_constraintTop_toBottomOf="@+id/edUsername" />

<Button
    android:id="@+id/btnLogin"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginTop="16dp"
    android:text="Login"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="@+id/edPassword"
    app:layout_constraintStart_toStartOf="@+id/edPassword"
    app:layout_constraintTop_toBottomOf="@+id/edPassword" />
</androidx.constraintlayout.widget.ConstraintLayout>

```

6. Buat *layout* antarmuka (UI) aplikasi dengan nama **activity_dashboard.xml** seperti berikut



Kode XML untuk antarmuka dalam **activity_dashboard.xml** sebagai berikut

```

<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/clDashboard"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <Button
        android:id="@+id/btnTombol"
        android:layout_width="wrap_content"

```

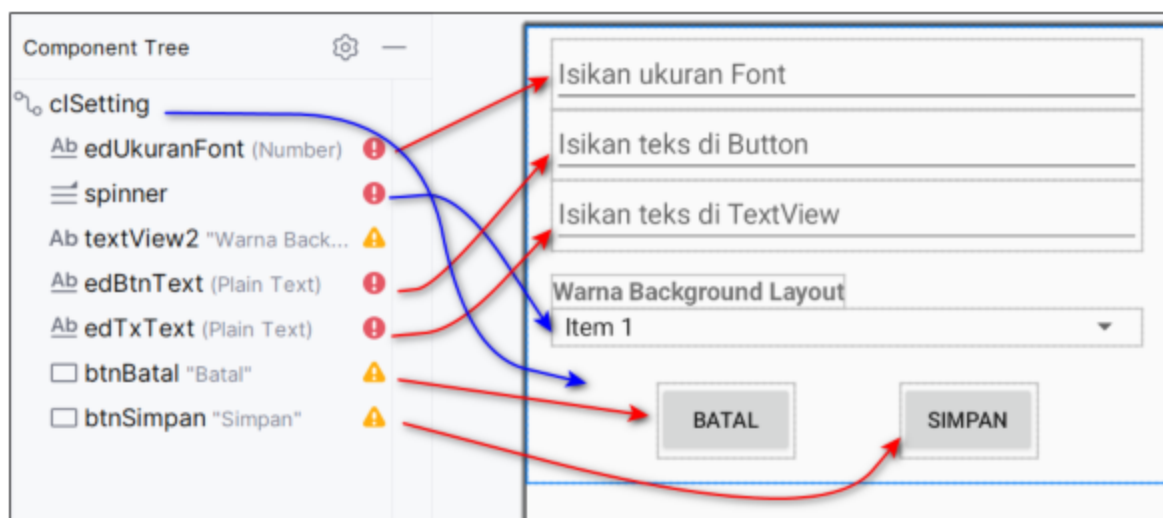
```

        android:layout_height="wrap_content"
        android:layout_marginTop="16dp"
        android:text="Tombol"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

<TextView
    android:id="@+id/txNarasi"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:layout_marginStart="16dp"
    android:layout_marginTop="16dp"
    android:layout_marginEnd="16dp"
    android:text="Teks Narasi"
    android:textAlignment="center"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toBottomOf="@+id/btnTombol" />
</androidx.constraintlayout.widget.ConstraintLayout>

```

7. Buat *layout* antarmuka (UI) aplikasi dengan nama **dialog_setting.xml** seperti berikut



Kode XML untuk antarmuka dalam **dialog_setting.xml** sebagai berikut

```

<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/clSetting"
    android:layout_width="match_parent"
    android:layout_height="wrap_content">

    <EditText
        android:id="@+id/edUkuranFont"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_marginStart="16dp"
        android:layout_marginTop="8dp"
        android:layout_marginEnd="16dp"
        android:ems="10"

```

```
        android:hint="Isikan ukuran Font"
        android:inputType="number"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

<Spinner
    android:id="@+id/spinner"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    app:layout_constraintEnd_toEndOf="@+id/edUkuranFont"
    app:layout_constraintHorizontal_bias="1.0"
    app:layout_constraintStart_toStartOf="@+id/edUkuranFont"
    app:layout_constraintTop_toBottomOf="@+id/textView2" />

<TextView
    android:id="@+id/textView2"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginTop="16dp"
    android:text="Warna Background Layout"
    android:textSize="16sp"
    android:textStyle="bold"
    app:layout_constraintStart_toStartOf="@+id/edTxText"
    app:layout_constraintTop_toBottomOf="@+id/edTxText" />

<EditText
    android:id="@+id/edBtnText"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:ems="10"
    android:hint="Isikan teks di Button"
    android:inputType="text"
    app:layout_constraintEnd_toEndOf="@+id/edUkuranFont"
    app:layout_constraintStart_toStartOf="@+id/edUkuranFont"
    app:layout_constraintTop_toBottomOf="@+id/edUkuranFont" />

<EditText
    android:id="@+id/edTxText"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:ems="10"
    android:hint="Isikan teks di TextView"
    android:inputType="text"
    app:layout_constraintEnd_toEndOf="@+id/edBtnText"
    app:layout_constraintStart_toStartOf="@+id/edBtnText"
    app:layout_constraintTop_toBottomOf="@+id/edBtnText" />

<Button
    android:id="@+id/btnBatal"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginStart="16dp"
    android:layout_marginTop="24dp"
    android:layout_marginBottom="16dp"
    android:text="Batal"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toStartOf="@+id/btnSimpan"
```

```

        app:layout_constraintHorizontal_bias="0.5"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/spinner" />

        <Button
            android:id="@+id/btnSimpan"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_marginEnd="16dp"
            android:text="Simpan"
            app:layout_constraintEnd_toEndOf="parent"
            app:layout_constraintHorizontal_bias="0.5"
            app:layout_constraintStart_toEndOf="@+id/btnBatal"
            app:layout_constraintTop_toTopOf="@+id/btnBatal" />
    </androidx.constraintlayout.widget.ConstraintLayout>

```

8. Buat **menu** dengan nama **menu_options.xml** yang akan digunakan untuk mengatur tampilan dashboard dengan kode xml sebagai berikut

```

<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android">

    <item
        android:id="@+id/mnSetting"
        android:title="Setting" />

</menu>

```

9. Tuliskan kode berikut ini pada file **MainActivity.kt**

```

package nugroho.benni.appl4

import android.content.Intent
import android.net.Uri
import android.os.Bundle
import android.view.View
import android.widget.Toast
import androidx.activity.enableEdgeToEdge
import androidx.appcompat.app.AppCompatActivity
import androidx.core.view.ViewCompat
import androidx.core.view.WindowInsetsCompat
import com.android.volley.DefaultRetryPolicy
import com.android.volley.Response
import com.android.volley.toolbox.StringRequest
import com.android.volley.toolbox.Volley
import nugroho.benni.appl4.databinding.ActivityMainBinding
import org.json.JSONObject

class MainActivity : AppCompatActivity(), View.OnClickListener {

    lateinit var b: ActivityMainBinding
    val url = Uri.parse(
        "http://192.168.20.239/kampus/login.php").toString()

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        b = ActivityMainBinding.inflate(layoutInflater)
        enableEdgeToEdge()
    }

```

```

        setContentView(b.root)
        ViewCompat.setOnApplyWindowInsetsListener(
            findViewById(b.root.id)) { v, insets ->
                val systemBars =
                    insets.getInsets(WindowInsetsCompat.Type.systemBars())
                v.setPadding(systemBars.left, systemBars.top,
                    systemBars.right, systemBars.bottom)
                insets
            }
        b.btnLogin.setOnClickListener(this)
    }

    //baca username dan password dari sharedPreferences
    //ketika aplikasi running
    override fun onStart() {
        super.onStart()
        readFromSharedPreferences()
    }

    //function simpan username dan password ke sharedPreferences
    fun writeToSharedPreferences() {
        val sharedPref =
            getSharedPreferences("LoginPref", MODE_PRIVATE) ?: return
        with(sharedPref.edit()) {
            putString("username", b.edUsername.text.toString())
            putString("password", b.edPassword.text.toString())
            apply()
        }
    }

    //function baca username dan password dari sharedPreferences
    fun readFromSharedPreferences() {
        //LoginPref adalah nama file untuk shared preference login
        val sharedPref =
            getSharedPreferences("LoginPref", MODE_PRIVATE) ?: return
        with(sharedPref) {
            //abc adalah nama user default
            b.edUsername.setText(getString("username", "abc"))
            //123 adalah password default
            b.edPassword.setText(getString("password", "123"))
        }
    }

    //function login
    fun login(username : String, password : String) {
        val request = object : StringRequest(Method.POST, url,
            Response.Listener {
                val jsonObject = JSONObject(it)
                val sukses = jsonObject.getString("sukses")
                if(sukses.equals("000")) {
                    writeToSharedPreferences()
                    val intent = Intent(
                        this, DashboardActivity::class.java)
                    startActivity(intent)
                } else Toast.makeText(this,
                    "Login gagal, username/password salah",
                    Toast.LENGTH_SHORT).show()
            },
    },

```

```

        Response.ErrorListener {
            Toast.makeText(this,
                "Login gagal, cek koneksi jaringan anda",
                Toast.LENGTH_SHORT).show()
        }
    ){
        override fun getParams(): MutableMap<String, String>? {
            val hm = HashMap<String, String>()
            hm.put("username", username)
            hm.put("password", password)
            return hm
        }
    }
    //setting timeout selama 5 detik
    request.setRetryPolicy(DefaultRetryPolicy(
        5000, //timeout 5 detik
        DefaultRetryPolicy.DEFAULT_MAX_RETRIES,
        DefaultRetryPolicy.DEFAULT_BACKOFF_MULT))

    val queue = Volley.newRequestQueue(this)
    queue.add(request)
}

override fun onClick(p0: View?) {
    login(b.edUsername.text.toString(), b.edPassword.text.toString())
}
}

```

10. Tuliskan kode berikut ini pada file **DashboardActivity.kt**

```

package nugroho.benni.appl4

import android.app.Dialog
import android.graphics.Color
import android.os.Bundle
import android.view.Menu
import android.view.MenuItem
import android.view.View
import android.view.ViewGroup
import android.widget.ArrayAdapter
import androidx.activity.enableEdgeToEdge
import androidx.appcompat.app.AppCompatActivity
import androidx.core.view.ViewCompat
import androidx.core.view.WindowInsetsCompat
import nugroho.benni.appl4.databinding.ActivityDashboardBinding
import nugroho.benni.appl4.databinding.DialogSettingBinding

class DashboardActivity : AppCompatActivity(), View.OnClickListener {
    lateinit var b: ActivityDashboardBinding
    lateinit var bDialog : DialogSettingBinding
    lateinit var dialog : Dialog
    lateinit var adapter : ArrayAdapter<String>
    val arrWarna = arrayOf(
        "Abu-abu", "Biru", "Cyan",
        "Hijau", "Hitam", "Kuning",
        "Merah", "Putih", "Ungu")

    //function baca data dari sharedPreferences

```

```

fun readFromSharedPreferences() {
    val sharedPref = getSharedPreferences("DashboardPref",
        MODE_PRIVATE) ?: return
    with(sharedPref) {
        b.btnTombol.text = getString("button", "Tombol")
        b.txNarasi.text = getString("textview", "Halo dunia")
        b.txNarasi.setTextSize(getFloat("fontsize", 16f))
        b.clDashboard.setBackgroundColor(getInt("background",
            Color.RED))
    }
}

//function menulis data ke sharedPreferences
fun writeToSharedPreferences() {
    val sharedPref = getSharedPreferences("DashboardPref",
        MODE_PRIVATE) ?: return
    with(sharedPref.edit()) {
        putString("button", bDialog.edBtnText.text.toString())
        putString("textview", bDialog.edTxText.text.toString())
        putFloat("fontsize",
            bDialog.edUkuranFont.text.toString().toFloat())

        val pos = adapter.getPosition(
            bDialog.spinner.selectedItem.toString())
        when(arrWarna[pos]) {
            "Merah" -> putInt("background", Color.RED)
            "Hijau" -> putInt("background", Color.GREEN)
            "Biru" -> putInt("background", Color.BLUE)
            "Kuning" -> putInt("background", Color.YELLOW)
            "Hitam" -> putInt("background", Color.BLACK)
            "Putih" -> putInt("background", Color.WHITE)
            "Abu-abu" -> putInt("background", Color.GRAY)
            "Ungu" -> putInt("background", Color.MAGENTA)
            "Cyan" -> putInt("background", Color.CYAN)
        }
        apply()
    }
}

override fun onCreate(savedInstanceState: Bundle?) {
    super.onCreate(savedInstanceState)
    b = ActivityDashboardBinding.inflate(layoutInflater)
    bDialog = DialogSettingBinding.inflate(layoutInflater)
    enableEdgeToEdge()
    setContentView(b.root)
    ViewCompat.setOnApplyWindowInsetsListener(
        findViewById(R.id.clDashboard)) { v, insets ->
        val systemBars = insets.getInsets(
            WindowInsetsCompat.Type.systemBars())
        v.setPadding(systemBars.left, systemBars.top,
            systemBars.right, systemBars.bottom)
        insets
    }
    bDialog.btnBatal.setOnClickListener(this)
    bDialog.btnSimpan.setOnClickListener(this)
    adapter = ArrayAdapter(this,
        android.R.layout.simple_spinner_item, arrWarna)
    bDialog.spinner.adapter = adapter
    dialog = Dialog(this)
}

```

```
}

override fun onStart() {
    super.onStart()
    readFromSharedPreferences()
}

override fun onOptionsItemSelected(item: MenuItem): Boolean {
    dialog.setContentView(bDialog.root)
    dialog.window?.setLayout(ViewGroup.LayoutParams.MATCH_PARENT,
        ViewGroup.LayoutParams.WRAP_CONTENT)
    dialog.setCancelable(false)
    dialog.show()
    return super.onOptionsItemSelected(item)
}

override fun onCreateOptionsMenu(menu: Menu?): Boolean {
    val mnuInflater = menuInflater
    mnuInflater.inflate(R.menu.menu_options, menu)
    return super.onCreateOptionsMenu(menu)
}

override fun onClick(p0: View?) {
    when(p0?.id) {
        bDialog.btnBatal.id ->{ dialog.dismiss() }
        bDialog.btnSimpan.id ->{
            writeToSharedPreferences()
            readFromSharedPreferences()
            dialog.dismiss()
        }
    }
}
}
```

D. Uji Coba Aplikasi

1. Pastikan antara web server dan aplikasi android dapat terhubung melalui jaringan LAN/internet
2. Jalankan aplikasi android, cobalah login ke aplikasi menggunakan username dan password yang terdaftar di database sehingga muncul halaman dashboard di aplikasi
3. Tampilan aplikasi ketika berjalan terlihat seperti gambar berikut ini

